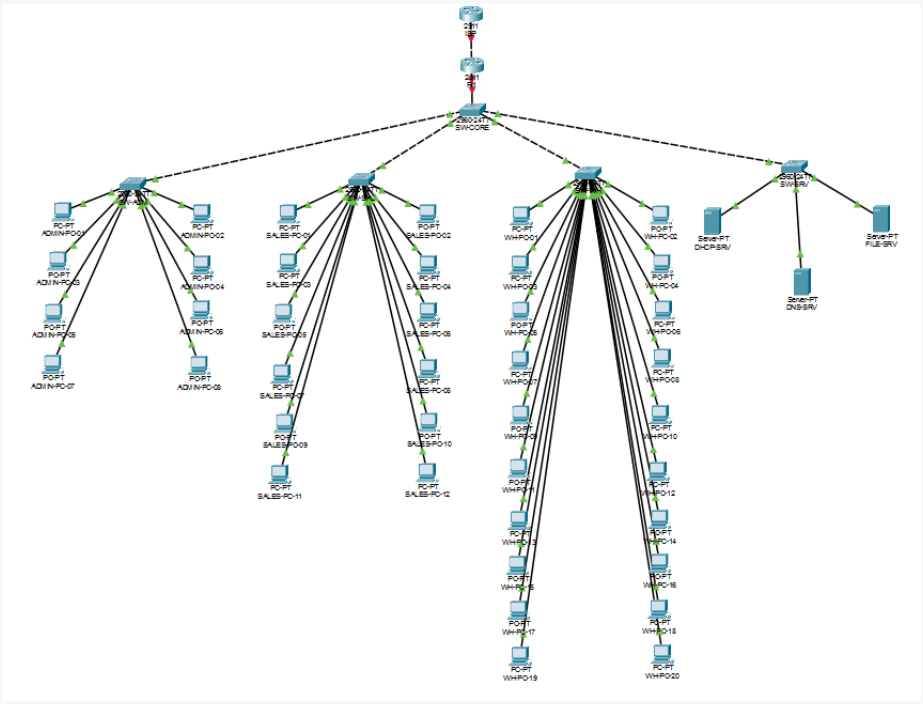
CMPG 325 · COMPUTER NETWORKS · NWU MAHIKENG

Technical Report

FINAL SUBMISSION · TSHIMO AGRI SUPPLIES NETWORK DESIGN & IMPLEMENTATION

Potchefstroom HQ, Agriculture | 10.23.0.0/16 | Fault Isolation — Advanced | 12 / 12 tests passed



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PROJECT ID CMPG325-2026-042	**CLIENT ID** CLI-042
ASSIGNED CLIENT Tshimo Agri Supplies	**ASSIGNED CHALLENGE** Network Troubleshooting

GitHub portfolio: github.com/MALULEKE-KS/tshimo-agri-network

CONTENTS

0 **Executive Summary**
The problem, the solution, the results, and where each Final deliverable lives

1 **Client & Requirements**
The brief, what was given, what was assumed, and how each requirement is traced to the build

2 **Design Rationale**
Topology and addressing decisions, the alternatives considered, and the arithmetic behind them

3 **Implementation**
Physical build, switching, routing and NAT, and the network services as configured in Packet Tracer

4 **Testing & Results**
Twelve functional tests run in the simulation, with the captured evidence

5 **Assigned Challenge — Advanced Fault Isolation**
A VLAN pruned from a trunk: baseline, symptoms, layered isolation, root cause, fix, verification

6 **Growth: Change Request 1 & the Branch Office**
CR1 absorbed on the live network; a proposed branch-office design that leaves HQ untouched

7 **Issues Encountered & Lessons**
The real problems hit during the build, how each was found and fixed

8 **Limitations & Future Work**
What the design does not do, and what would be added next

9 **Reflection**
What the project taught, and what would be done differently

10 **GitHub Portfolio**
Repository structure, evidence index and commit history

A **Appendix — Device Configurations**
The IOS command sets applied to every switch and router, verbatim from the repository

NETWORK DEVICES	END DEVICES	PHYSICAL LINKS
7	40 PCs + 3 servers	49

VLANS	FUNCTIONAL TESTS	ASSIGNED FAULT
10 · 20 · 30 · 40 · 99	12 / 12 passed	Isolated & repaired

HOW THIS REPORT IS WRITTEN

Every result in this report comes from the running Packet Tracer network: a screenshot or a pasted command output. Where something is a design proposal that was **not** built or simulated (the branch-office design in Section 6.2, the fault catalogue in Section 5.9), it is labelled as a proposal. Where a result was less clean than it could have been, the report says so (Sections 4.2 and 7).

0 EXECUTIVE SUMMARY

Final Submission · 16 October 2026

The problem

Tshimo Agri Supplies, an agricultural supplier in Potchefstroom, needs a head-office network: separated departments, internal DHCP, DNS and file services, and internet access over one ISP link, inside the assigned block `10.23.0.0/16`. A **branch office may open within 18 months** and **Change Request CR1** adds eight staff to one department; neither may force a redesign. The assigned challenge is **Network Troubleshooting — fault isolation, Advanced**.

The solution

One edge router (R1) does router-on-a-stick inter-VLAN routing, DHCP relay and NAT/PAT. One Layer 2 core switch trunks R1 to four access switches, one per department. Four business VLANs, each a uniform `/26` (62 usable hosts) whatever the current headcount, plus management/native VLAN 99. All of HQ sits inside `10.23.0.0/23`, one summarisable block, and `10.23.128.0/17` is reserved untouched for the branch.

The results

1. The design was built in Cisco Packet Tracer 9.0.1 **without changing a single VLAN, subnet or gateway** from the Milestone 1 plan: 7 network devices, 40 PCs, 3 servers, 49 links.
2. **12 of 12 functional tests passed**: cabling, VLANs, trunks, router interfaces and routes, DHCP via relay, same-VLAN and inter-VLAN pings, DNS and web, NAT, and CR1 capacity.
3. **CR1 was absorbed on the live network**: Warehouse grew from 12 to 20 PCs by assigning eight more access ports to VLAN 30. No subnet, gateway, trunk or router change was needed.
4. **The assigned fault was injected, isolated and fixed.** VLAN 30 was removed from one trunk; Warehouse lost its gateway while local switching survived; an Admin control test scoped it; `show interfaces trunk` isolated it; a one-line fix and repeat tests verified recovery.
5. The branch requirement is met by design: a proposed plan (Section 6.2) needs only a spare router port, one static route and one ACL line at HQ.

Where each Final deliverable is

FINAL DELIVERABLE	DELIVERED AS	LOCATION
Packet Tracer project (.pkt)	Complete HQ network, repaired and healthy	GitHub `packet-tracer/` and `submissions/final/`; link in the eFundi text
GitHub portfolio of evidence	Docs, configs, 20 screenshots, test log, fault record, three frozen packages	github.com/MALULEKE-KS/tshimo-agri-network
Technical report	This document	**`Final_TechnicalReport_Maluleke_48277444.pdf`**
15–20 minute inset video	Recorded walkthrough and live demonstration	Link in the eFundi text (not in the repository)

> **Relationship to earlier submissions.** This report builds on my own Milestone 1 and 2 submissions for the same project. Carried forward from Milestone 2: the evidence figures, test results, fault demonstration and configurations. New here: the design rationale (Section 2), the fault-method table (5.9), the branch proposal (6.2), limitations (Section 8) and reflection (Section 9).

1 CLIENT & REQUIREMENTS

1.1 What the brief gives

ITEM	VALUE
Client	Tshimo Agri Supplies, Potchefstroom — agriculture industry (Client ID CLI-042)
Assigned address block	`10.23.0.0/16`
Assigned challenge	Network Troubleshooting — fault isolation scenario, Advanced difficulty
Branch office	May open within 18 months; the addressing plan must allow it without redesigning HQ
Change Request CR1	Eight additional staff in one department; capacity must be added without redesigning the network

1.2 What the brief does not give — and what was assumed

The brief does not describe Tshimo Agri Supplies' departments or headcount. The structure below was **assumed** at Milestone 1 for a small-to-medium agri-supply distributor, and everything after it (VLANs, subnets, port counts, DHCP pools) was derived from it. It is an assumption, not a client fact; a different real headcount would change the numbers but not the method.

DEPARTMENT	HEADCOUNT	RATIONALE FOR THE ASSUMPTION
Admin / Management	8	Small office function: owner, finance, HR
Sales	12	Client-facing and order-desk staff
Warehouse / Logistics	12 → 20 (CR1)	Largest department: stock handling, dispatch, delivery; the department CR1 grows, as the most plausible place for an agri distributor to add headcount
Servers	3	DHCP, DNS, and file/web services hosted on-premises

1.3 Requirements traced to the build

#	REQUIREMENT	HOW THE DESIGN MEETS IT	EVIDENCE
R1	Department segmentation	One VLAN and one `/26` per department; inter-VLAN traffic must pass through R1	Sections 3.2, 4 · Fig 4, 11
R2	Growth headroom (CR1)	Every VLAN is a `/26` with 62 usable hosts; 24-port access switches. Warehouse went from 12 to 20 with no subnet change	Section 6.1 · Fig 4
R3	Branch office without redesign	`10.23.128.0/17` reserved and unused; HQ summarises as `10.23.0.0/23`	Sections 2.3, 6.2 · Fig 1
R4	Internet through one ISP link	Default route to the ISP; NAT/PAT overload on R1's WAN address	Section 3.3 · Fig 6, 10
R5	Internal DHCP, DNS, file services	Three Server-PT servers in VLAN 40; DHCP relayed by R1; DNS A record; HTTP and FTP	Section 3.4 · Fig 7, 8
R6	Support an Advanced troubleshooting exercise	Multiple VLANs, trunks and a routed uplink give a real fault space; the VLAN-pruning fault was injected and isolated	Section 5 · Fig 12–16

2 DESIGN RATIONALE

Each major decision, what else was possible, and why this option won for a client of this size.

2.1 Topology and service decisions

DECISION	ALTERNATIVE	WHY THIS OPTION	TRADE-OFF ACCEPTED
Router-on-a-stick on R1	Layer 3 switch with SVIs	All routing, DHCP relay and NAT sit in one device, which keeps the fault space small and the isolation method clear. It is also the technique the course emphasises, and a 2960 is Layer 2 only.	One 1 Gbps trunk carries all inter-VLAN traffic; ample for 43 hosts
One Layer 2 core switch	Redundant dual-core	Fewer hops to reason about; matches the client's size	SW-CORE and R1 are single points of failure (Section 8)
One access switch per department	One large switch with port-based VLANs	Physical separation mirrors the logical one; 24-port switches give port headroom for CR1	Four switches to manage instead of one
Dedicated DHCP server in VLAN 40, relayed by R1	DHCP pools on R1 itself	Central, GUI-managed pools; a server VLAN is where a business would host it; demonstrates relay (`ip helper-address`)	New leases depend on VLAN 40, the relay and the server all being up
Static addresses for servers	DHCP reservations	Servers must be at predictable addresses (DNS and DHCP clients point at them)	Manual change if a server is re-addressed
NAT/PAT overload to R1's WAN address	Static or pooled NAT	The client has no public allocation, so every host shares one address, told apart by port	No inbound connections without extra port-forwarding
Native VLAN 99, never VLAN 1	Default native VLAN 1	Keeps untagged traffic out of the default VLAN; a native mismatch also logs loudly, which helps troubleshooting	Native VLAN must be set on both ends of every trunk
Static 802.1Q trunks, explicit allowed list	Auto-negotiated, all VLANs allowed	Predictable behaviour; only the five VLANs the design uses may cross a trunk. This is also what makes the assigned fault possible	A wrong list silently drops a VLAN (the assigned fault)

2.2 The addressing arithmetic

A `/26` leaves 6 host bits: 2^6 = 64 addresses, minus the network and broadcast addresses = **62 usable hosts**. Each business VLAN gets one, whatever its headcount today:

VLAN	SUBNET	USABLE RANGE	DEVICES NOW	USED OF 62	FREE
10 ADMIN	`10.23.0.0/26`	`.1 – .62`	8 PCs	13%	54
20 SALES	`10.23.0.64/26`	`.65 – .126`	12 PCs	19%	50
30 WAREHOUSE	`10.23.0.128/26`	`.129 – .190`	20 PCs (12 + 8 CR1)	32%	42
40 SERVERS	`10.23.0.192/26`	`.193 – .254`	3 servers	5%	59
99 MGMT-NATIVE	`10.23.1.16/28`	`.17 – .30`	gateway + 5 switch SVIs	43% of 14	8

Usable ranges include the R1 gateway address (the first usable address of each subnet); "Free" subtracts the gateway and devices. **Why uniform and over-sized?** A `/27` (30 hosts) would also have covered CR1's 20 Warehouse staff, but would leave only 10 spare, so the next growth step would force a re-address. The four business `/26`s together use 256 addresses, 0.4% of the `/16`; the whole HQ footprint, `10.23.0.0/23`, is 512 addresses, 0.78%. Over-provisioning costs almost nothing here, and one uniform rule (every

department gets a `/26`) is easier to operate and to explain than four custom sizes. A real limit to note: the DHCP pools lease 50 addresses, so the first ceiling Warehouse would hit is the pool (raisable to 53 from its current start of `.138`) rather than the subnet.

2.3 Address-space map and summarisation

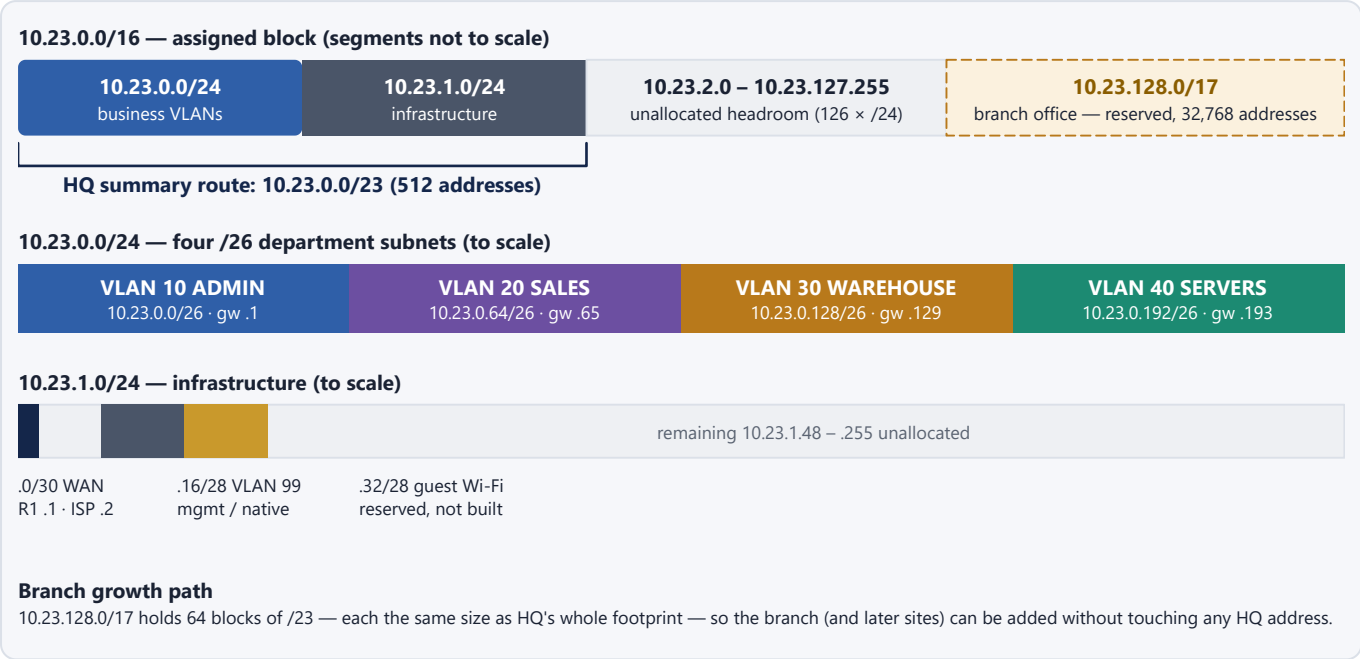


Figure 1. *The assigned `/16`. HQ's business VLANs and infrastructure occupy two contiguous `/24`s, so they summarise as one `10.23.0.0/23` route. The upper half of the `/16` is held back for the branch.*

Why the summary works. `10.23.0.0/24` and `10.23.1.0/24` differ only in the last bit of the third octet, so the first 23 bits are identical and one `/23` covers both. Any future router upstream of HQ needs one route, not nine. **Why a whole `/17` for the branch?** The brief gives no branch headcount or structure, so rather than invent one, the plan fixes what is known (HQ) and holds back half the allocation, untouched, until the branch's real requirements exist.

3 IMPLEMENTATION

The design was built in Cisco Packet Tracer 9.0.1. Every device configuration was written as a reviewable command file first, applied one device at a time, and checked with a `show` command before moving on. The full command sets are in the Appendix.

3.1 Physical build and cabling

DEVICE	MODEL	ROLE
ISP, R1	Cisco 2911	ISP: simulated provider with loopback `203.0.113.1` as the "internet" target. R1: edge router (routing, relay, NAT)
SW-CORE	Cisco 2960-24TT	Layer 2 distribution; trunks R1 to all four access switches
SW-ADM / SAL / WH / SRV	Cisco 2960-24TT	One access switch per department; 24 ports each for physical headroom
40 PCs, 3 servers	PC-PT, Server-PT	8 Admin, 12 Sales, 20 Warehouse (12 + 8 CR1); DHCP-SRV, DNS-SRV, FILE-SRV

LINK	PORTS	CABLE	ROLE
ISP ↔ R1	**ISP Gi0/0 ↔ R1 Gi0/1**	Crossover	Routed WAN `10.23.1.0/30`
R1 ↔ SW-CORE	**R1 Gi0/0 ↔ SW-CORE Gi0/1**	Straight-through	802.1Q trunk
SW-CORE ↔ access	**Fa0/1→SW-ADM, Fa0/2→SW-SAL, Fa0/3→SW-WH, Fa0/4→SW-SRV (each to Fa0/1)**	Crossover	802.1Q trunks
Access ↔ endpoints	**Fa0/2 upward**	Straight-through	Access ports

Cabling follows the standard rule: straight-through between unlike devices, crossover between like devices. The 2911 and 2960 would usually bring a wrong cable up through Auto-MDIX, but the build uses the textbook-correct cable so it does not rely on that.

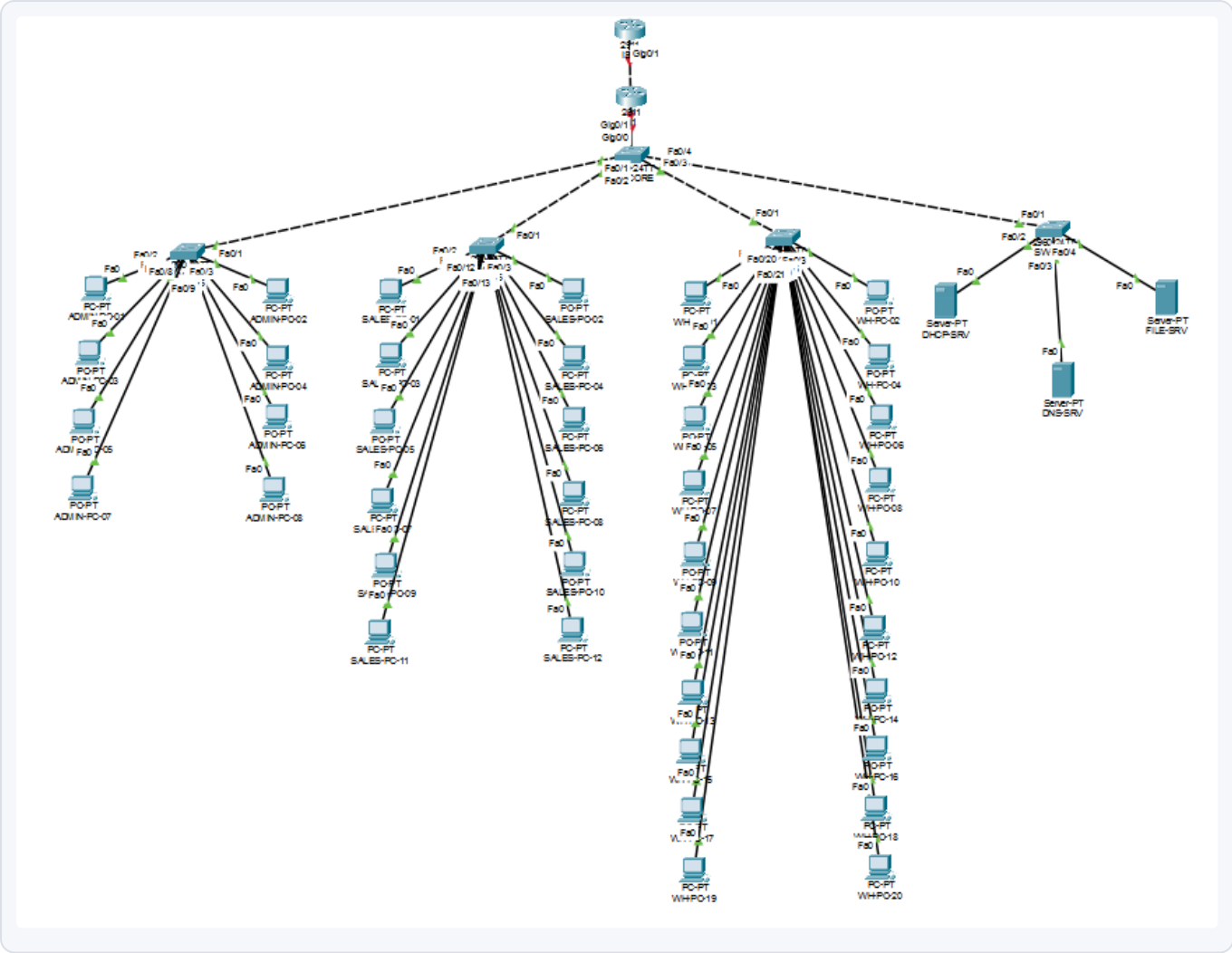


Figure 2. *The implemented HQ topology with port labels on: ISP → R1 → SW-CORE → four access switches → 40 PCs and 3 servers. The four SW-CORE ↔ access links are dashed (crossover); endpoint links are solid (straight-through). The short ISP–R1 link is too small at this zoom to show its dash pattern.*

3.2 VLANs and trunks

1. **VLAN database.** SW-CORE holds all five VLANs because every VLAN crosses it. Each access switch creates only the VLAN its ports use, plus VLAN 99.
2. **Trunks.** Every switch-to-switch and switch-to-router link is a static 802.1Q trunk (`switchport mode trunk`) with **native VLAN 99** and an explicit allowed list of `10,20,30,40,99`.
3. **Access ports.** Each department's port range is a static access port in its VLAN, with PortFast so PCs come up without waiting for spanning tree.
4. **Management.** Each switch has an SVI on VLAN 99 — SW-CORE `.18`, SW-ADM `.19`, SW-SAL `.20`, SW-WH `.21`, SW-SRV `.22` (all `/28`) — with default gateway `10.23.1.17`.

```
SW-CORE#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      99
Fa0/2       on           802.1q         trunking      99
Fa0/3       on           802.1q         trunking      99
Fa0/4       on           802.1q         trunking      99

Port        Vlans allowed on trunk
Fa0/1       10,20,30,40,99
Fa0/2       10,20,30,40,99
Fa0/3       10,20,30,40,99
Fa0/4       10,20,30,40,99

Port        Vlans allowed and active in management domain
Fa0/1       10,20,30,40,99
Fa0/2       10,20,30,40,99
Fa0/3       10,20,30,40,99
Fa0/4       10,20,30,40,99

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       10,20,30,40,99
Fa0/2       10,20,30,40
Fa0/3       10,20,30,40
 --More--
```

Figure 3. *SW-CORE `show interfaces trunk`: Fa0/1–Fa0/4 trunking, 802.1Q, native VLAN 99, allowed `10,20,30,40,99`. (Gi0/1 to R1 had not yet come up when this was captured, because R1 was configured afterwards.)*

```
SW-WH#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                                Gig0/2
30   WAREHOUSE                        active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
99   MGMT-NATIVE                      active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SW-WH#
```

Figure 4. *SW-WH `show vlan brief`: VLAN 30 WAREHOUSE holds exactly 20 access ports, Fa0/2–Fa0/21 — the 12 original Warehouse staff plus the 8 from CR1.*

3.3 Routing, WAN and NAT (R1)

The switches are Layer 2 only, so R1 is the one place traffic moves between VLANs. Its Gi0/0 carries no IP address; instead it has one 802.1Q subinterface per VLAN, and each subinterface's address is that VLAN's default gateway.

SUBINTERFACE	ENCAPSULATION	IP ADDRESS	DHCP RELAY	NAT
`Gi0/0.10`	`dot1Q 10`	`10.23.0.1/26`	`10.23.0.194`	inside
`Gi0/0.20`	`dot1Q 20`	`10.23.0.65/26`	`10.23.0.194`	inside
`Gi0/0.30`	`dot1Q 30`	`10.23.0.129/26`	`10.23.0.194`	inside
`Gi0/0.40`	`dot1Q 40`	`10.23.0.193/26`	— (servers are static)	inside
`Gi0/0.99`	`dot1Q 99 native`	`10.23.1.17/28`	—	—
`Gi0/1`	— (WAN)	`10.23.1.1/30`	—	outside

1. **Default route:** `ip route 0.0.0.0 0.0.0.0 10.23.1.2` sends everything not local to the ISP.
2. **NAT/PAT overload:** `access-list 1 permit 10.23.0.0 0.0.0.255` with `ip nat inside source list 1 interface Gi0/1 overload`. Every user and server shares R1's WAN address `10.23.1.1`, told apart by port number.
3. **Deliberately not translated:** the ACL matches only `10.23.0.0/24`. The WAN and management ranges (`10.23.1.0/24`) keep their real addresses. The ISP needs no return route to the LAN, because it only ever sees R1's translated address.

```
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.10  10.23.0.1       YES manual up                    up
GigabitEthernet0/0.20  10.23.0.65      YES manual up                    up
GigabitEthernet0/0.30  10.23.0.129     YES manual up                    up
GigabitEthernet0/0.40  10.23.0.193     YES manual up                    up
GigabitEthernet0/0.99  10.23.1.17      YES manual up                    up
GigabitEthernet0/1     10.23.1.1       YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R1#
```

Figure 5. *R1 `show ip interface brief`: all five subinterfaces and the WAN link up/up at their planned addresses.*

```
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.10  10.23.0.1       YES manual up                    up
GigabitEthernet0/0.20  10.23.0.65      YES manual up                    up
GigabitEthernet0/0.30  10.23.0.129     YES manual up                    up
GigabitEthernet0/0.40  10.23.0.193     YES manual up                    up
GigabitEthernet0/0.99  10.23.1.17      YES manual up                    up
GigabitEthernet0/1     10.23.1.1       YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R1#how ip route
       ^
% Invalid input detected at '^' marker.

R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.23.1.2 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 12 subnets, 4 masks
C       10.23.0.0/26 is directly connected, GigabitEthernet0/0.10
L       10.23.0.1/32 is directly connected, GigabitEthernet0/0.10
C       10.23.0.64/26 is directly connected, GigabitEthernet0/0.20
L       10.23.0.65/32 is directly connected, GigabitEthernet0/0.20
C       10.23.0.128/26 is directly connected, GigabitEthernet0/0.30
L       10.23.0.129/32 is directly connected, GigabitEthernet0/0.30
C       10.23.0.192/26 is directly connected, GigabitEthernet0/0.40
L       10.23.0.193/32 is directly connected, GigabitEthernet0/0.40
C       10.23.1.0/30 is directly connected, GigabitEthernet0/1
L       10.23.1.1/32 is directly connected, GigabitEthernet0/1
C       10.23.1.16/28 is directly connected, GigabitEthernet0/0.99
 --More--
```

Figure 6. *R1 `show ip route`: connected and local routes for every VLAN and the WAN, and "Gateway of last resort is 10.23.1.2". (The one "% Invalid input" line is a typo, `how` for `show`, corrected on the next line.)*

3.4 Servers, DHCP with relay, and DNS

SERVER	ADDRESS (VLAN 40)	SERVICE
DHCP-SRV	**10.23.0.194/26**	DHCP: pools ADMIN `.10`, SALES `.74`, WH `.138` (50 leases each); server pool `.200` (20)
DNS-SRV	**10.23.0.195/26**	DNS: `files.tshimo.test → 10.23.0.196`
FILE-SRV	**10.23.0.196/26**	HTTP and FTP

A PC's DHCP request is a broadcast that never leaves its own VLAN. R1's `ip helper-address 10.23.0.194` on each user subinterface forwards it to DHCP-SRV as unicast, stamped with the subinterface's gateway address; the server uses that address to choose the matching pool. Each pool starts nine addresses above its gateway, which keeps `.1–.9` of every `/26` free for infrastructure. All 40 PCs were set to DHCP and each department's first lease was checked. Sales and Warehouse are shown below; Admin's lease (ADMIN-PC-01 received `10.23.0.10/26`, gateway `.1`) is in the repository as `screenshots/06a-admin-dhcp.png`.

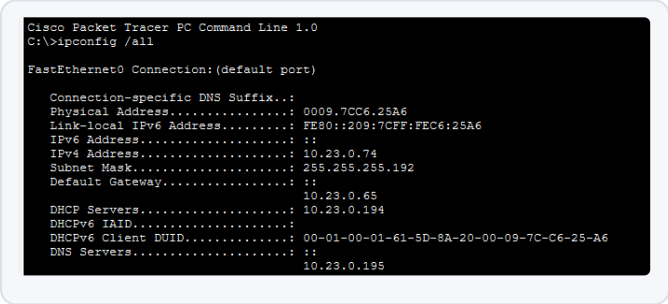


Figure 7a. *SALES-PC-01: `10.23.0.74/26`, gateway `.65`, DHCP `.194`, DNS `.195`.*

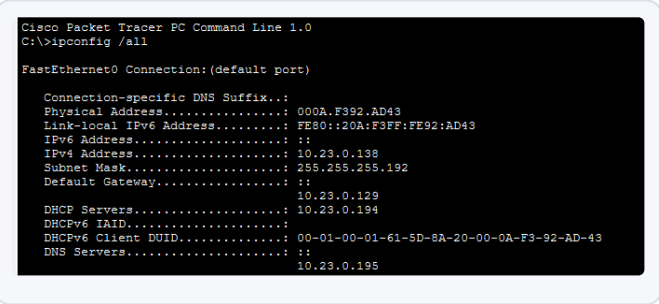


Figure 7b. *WH-PC-01: `10.23.0.138/26`, gateway `.129`, DHCP `.194`, DNS `.195`.*

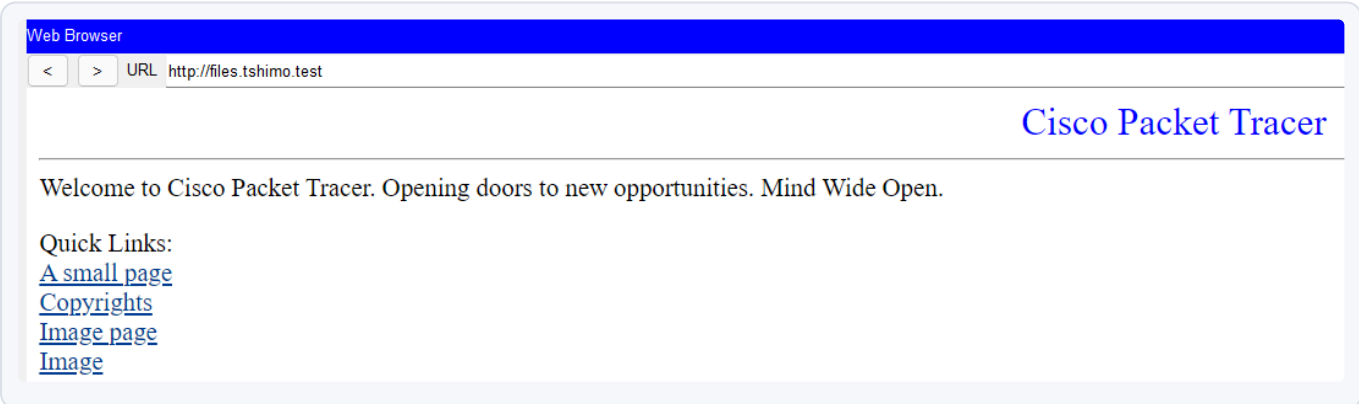


Figure 8. *ADMIN-PC-04's browser opening `http://files.tshimo.test`: DNS-SRV resolves the name and FILE-SRV serves the page.*

4 TESTING & RESULTS

Every test was run live in Packet Tracer against the configured network. "Observed" is recorded only from what the simulation showed.

4.1 Functional test results

ID	TEST	OBSERVED	RESULT	FIG.
PHY-01	Topology and cabling	7 network devices, 43 end devices, 49 links; crossover on like-device links	**PASS**	2
L2-00	SW-CORE `show cdp neighbors`	Fa0/1–Fa0/4 → SW-ADM, SW-SAL, SW-WH, SW-SRV as planned	**PASS**	repo
L2-01	SW-WH `show vlan brief`	VLAN 30 active on Fa0/2–Fa0/21	**PASS**	4
L2-02	SW-CORE `show interfaces trunk`	Trunks: 802.1Q, native 99, allowed 10,20,30,40,99	**PASS**	3
L3-01	R1 `show ip interface brief`	5 subinterfaces + Gi0/1 up/up at planned addresses	**PASS**	5
L3-02	R1 `show ip route`	All VLAN and WAN routes connected; default via 10.23.1.2	**PASS**	6
DHCP-01	One PC per user VLAN	Each took the first address of its own pool, with correct gateway and DNS	**PASS**	7a, 7b, repo
LAN-01	WH-PC-01 → WH-PC-02	4/4 replies (same VLAN)	**PASS**	9
ROUTE-01	Admin and Sales → FILE-SRV	ADMIN-PC-01 3/4 (first lost to ARP), SALES-PC-01 4/4	**PASS**	11
DNS-01	Browser → `files.tshimo.test`	Name resolved, page loaded	**PASS**	8
NAT-01	PC → `203.0.113.1`	SALES-PC-03 4/4, 0% loss	**PASS**	10
CR1-01	Warehouse capacity	20 ports in VLAN 30; no subnet, gateway or trunk change	**PASS**	4

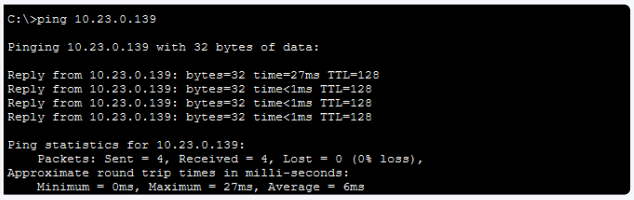


Figure 9. *WH-PC-01 → WH-PC-02 (`10.23.0.139`), same VLAN: 4/4.*

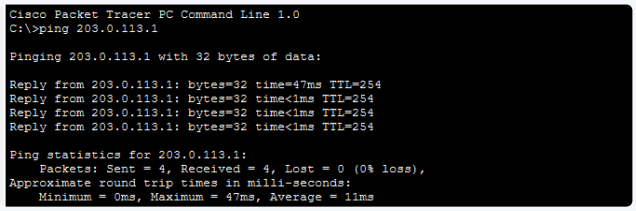


Figure 10. *SALES-PC-03 → `203.0.113.1` (ISP loopback) through NAT: 4/4, 0% loss.*

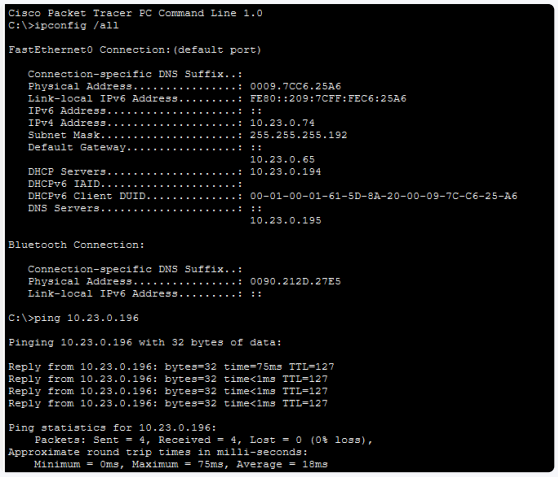


Figure 11. *SALES-PC-01 (VLAN 20, `10.23.0.74`) → FILE-SRV (VLAN 40, `10.23.0.196`): 4/4. TTL 127 instead of 128 shows the packet crossed one router hop (R1).*

4.2 Where the evidence is weaker than ideal

1. **NAT table.** `show ip nat translations` was run about six minutes after the NAT ping and returned empty, because dynamic PAT entries expire when idle. The successful ping in Figure 10 is the proof (an untranslated `10.23.0.x` source could not get a reply from the ISP), but the better practice is to run the table check immediately after the ping.

2. **Warehouse inter-VLAN.** Warehouse's own ping to the file server appears in the fault baseline (Figure 12, 4/4), not as a standalone ROUTE-01 capture.

3. **ISP–R1 crossover.** The link is too short at screenshot zoom to show its dash pattern. The link works (the NAT ping crosses it), and the cable was placed as crossover per the cable schedule.

5 ADVANCED FAULT ISOLATION

5.1 The scenario

A change on the core switch removes VLAN 30 (Warehouse) from the allowed-VLAN list on the SW-CORE → SW-WH trunk. It is a realistic one-word mistake: typing `switchport trunk allowed vlan 10,20,40,99` instead of `…10,20,30,40,99`, or running `remove 30` on the wrong port.

Why it qualifies as Advanced:

1. The physical link stays up. Link lights and Layer 1 checks look healthy, so the fault cannot be seen on the topology.
2. The failure is partial. Warehouse PCs can still reach each other, which can mislead a troubleshooter into blaming the gateway, DHCP or the PCs.
3. The two ends of the trunk disagree: SW-WH still allows VLAN 30 and SW-CORE does not. Checking only one end misses it.
4. It is a one-line mistake with a one-line fix, so finding it depends on method, not on rebuilding anything.

5.2 Method

Work up the OSI model and move up a layer only once the one below is proven good. Use a **control test** from an unaffected department to scope the fault before touching any device. Treat expected symptoms as hypotheses to check, not conclusions.

5.3 Step 1 — Healthy baseline

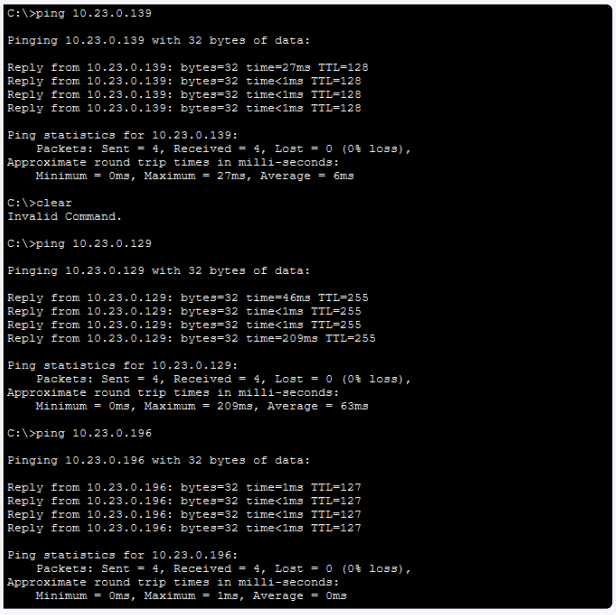


Figure 12. *Before the fault, WH-PC-01 (`10.23.0.138`) reaches WH-PC-02 (`.139`), its gateway (`.129`) and FILE-SRV (`.196`): all 4/4. The Packet Tracer file was saved at this point.*

5.4 Step 2 — Inject the fault (on SW-CORE, not saved)

```
configure terminal
interface fastEthernet0/3
 switchport trunk allowed vlan remove 30
end           ! no write memory — the fault is never saved
```

5.5 Step 3 — Observe the symptoms

TEST	EXPECTED (HYPOTHESIS)	OBSERVED
WH-PC-01 → gateway `.129`	Fail: the path to R1 crosses the trunk	100% loss
WH-PC-01 → FILE-SRV `.196`	Fail: another VLAN, needs R1	100% loss
WH-PC-01 → WH-PC-02 `.139`	Work: same switch, same VLAN, no trunk needed	4/4
ADMIN-PC-01 → FILE-SRV (control)	Work: Admin uses Fa0/1, not Fa0/3	4/4

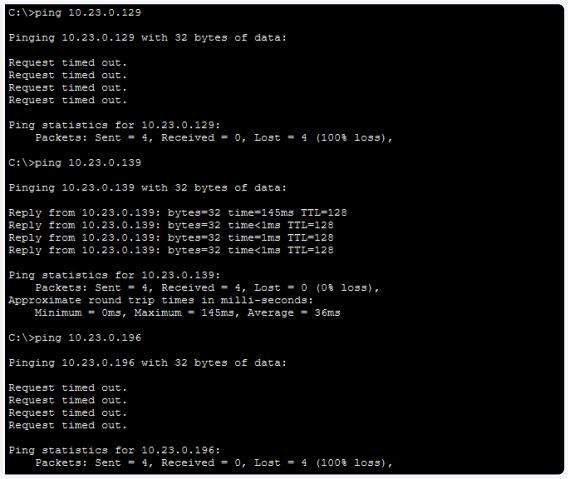


Figure 13. *WH-PC-01 during the fault: gateway (`.129`) and file server (`.196`) fail; WH-PC-02 (`.139`) still answers.*

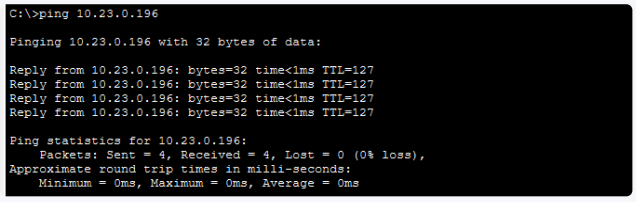


Figure 14. *ADMIN-PC-01 control test during the same fault: FILE-SRV answers 4/4, 0% loss.*

What the two results prove together. The control test passes and Warehouse fails, so the fault is scoped to **Warehouse only**. Everything Admin depends on (R1's routing, SW-CORE's other ports, SW-SRV, the file server) works. What is left to suspect is the path only Warehouse uses: SW-WH, its uplink to SW-CORE, or R1's own VLAN 30 subinterface.

5.6 Step 4 — Isolate layer by layer

LAYER	EVIDENCE	CONCLUSION
Scope	Admin works; Warehouse PCs reach each other	The servers and SW-WH's access ports are fine. The problem is on the path between SW-WH and R1.
1 — Physical	Link lights up; WH-PC-01 ↔ WH-PC-02 works	Cables and ports are fine.
3 — Network	WH-PC-01 still holds its lease `10.23.0.138/26`, gateway `.129`; R1's Gi0/0.30 was up/up (Figure 5)	Not a host addressing or gateway-configuration fault. (R1's subinterface was not re-checked during the fault; the fix on SW-CORE alone restoring service confirms R1 was never the cause.)
2 — Data link	SW-CORE `show interfaces trunk` (Figure 15)	All five trunk ports up, native 99. Fa0/3's allowed list reads `10,20,40,99`; every other port reads `10,20,30,40,99`. **Fault found.**

Root cause. SW-CORE Fa0/3 is the only physical path between SW-WH and the rest of the network. With VLAN 30 removed from that trunk's allowed list, no VLAN 30 frame can cross it. Warehouse PCs can still switch locally, but lose everything that depends on reaching R1. Mode, encapsulation and native VLAN are all correct, so the fault is purely the trunk's VLAN membership.

```
SW-CORE#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW-CORE(config)#interface fastEthernet0/3
SW-CORE(config-if)# switchport trunk allowed vlan remove 30
SW-CORE(config-if)#end
SW-CORE#
%SYS-5-CONFIG_I: Configured from console by console

SW-CORE#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW-CORE(config)#interface fastEthernet0/3
SW-CORE(config-if)# switchport trunk allowed vlan remove 30
SW-CORE(config-if)#end
SW-CORE#
%SYS-5-CONFIG_I: Configured from console by console
configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW-CORE(config)#interface fastEthernet0/3
SW-CORE(config-if)# switchport trunk allowed vlan remove 30
SW-CORE(config-if)#end
SW-CORE#
%SYS-5-CONFIG_I: Configured from console by console

SW-CORE#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      99
Fa0/2       on           802.1q         trunking      99
Fa0/3       on           802.1q         trunking      99
Fa0/4       on           802.1q         trunking      99
Gig0/1      on           802.1q         trunking      99

Port        Vlans allowed on trunk
Fa0/1       10,20,30,40,99
Fa0/2       10,20,30,40,99
Fa0/3       10,20,40,99
Fa0/4       10,20,30,40,99
Gig0/1      10,20,30,40,99

Port        Vlans allowed and active in management domain
Fa0/1       10,20,30,40,99
Fa0/2       10,20,30,40,99
Fa0/3       10,20,40,99
Fa0/4       10,20,30,40,99
Gig0/1      10,20,30,40,99

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       10,20,30,40,99
Fa0/2       10,20,30,40,99
Fa0/3       10,20,40,99
Fa0/4       10,20,30,40,99
Gig0/1      10,20,30,40,99
```

Figure 15. *The key evidence: SW-CORE `show interfaces trunk`. Fa0/3 alone is missing VLAN 30 across the allowed, active and forwarding lists. Gi0/1 to R1 is trunking normally.*

5.7 Step 5 — Apply the smallest fix (and save it)

```
configure terminal
interface fastEthernet0/3
 switchport trunk allowed vlan add 30
end
write memory  ! [OK] — the fix is saved; the fault never was
```

`add 30` restores exactly the one VLAN that was removed, without retyping, and so without risking, the rest of the list.

5.8 Step 6 — Verify by repeating the failed tests

CHECK	EXPECTED	OBSERVED
SW-CORE `show interfaces trunk`	Fa0/3 allows 10,20,30,40,99, as the others	**PASS** Fa0/3 back to `10,20,30,40,99`
WH-PC-01 → gateway `.129`	Replies	**4/4**
WH-PC-01 → FILE-SRV `.196`	Replies	**4/4**

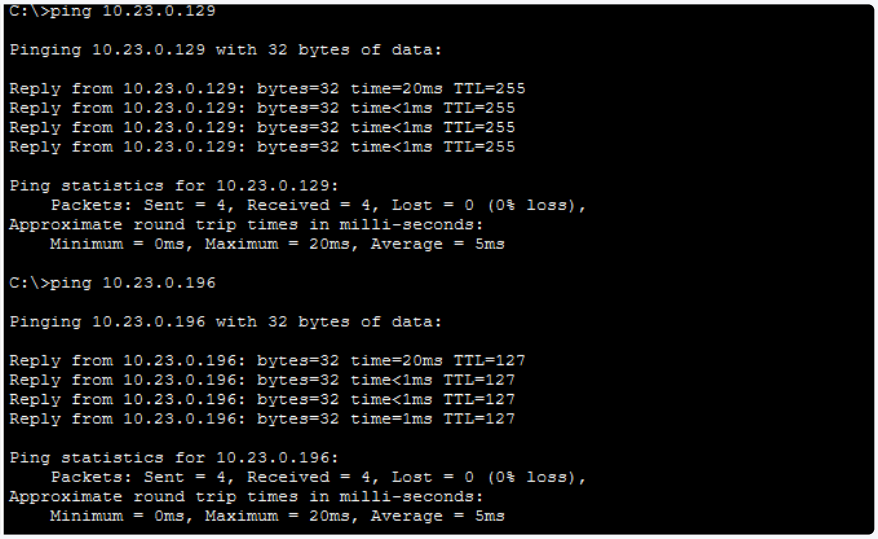


Figure 16. *After the fix: WH-PC-01 reaches its gateway (`.129`) and FILE-SRV (`.196`) again, 4/4 each, 0% loss.*

> **Final state.** The submitted `.pkt` was saved after the fix, with `write memory` run on SW-CORE, so it contains the healthy network, not the faulted one.

5.9 The same method applied to other faults

analysis — not injected

Only the VLAN-pruning fault was injected and tested. The table below is analysis of how the same scoped, layer-by-layer method separates other plausible faults in *this* network. It is reasoning, not observed results. It shows why the control test and the choice of first command matter: several of these produce a symptom close to the one demonstrated.

FAULT	SYMPTOM PATTERN	FIRST DISCRIMINATING CHECK	CONFIRMS WITH
VLAN pruned from trunk (done)	One VLAN loses its gateway and other VLANs; local switching works; other VLANs fine	Control test from another department	`show interfaces trunk` on both ends: one allowed list differs
Native VLAN mismatch	Console logs `NATIVE_VLAN_MISMATCH`; STP may block the VLAN on that port; management traffic on the trunk fails	Read the syslog messages	`show interfaces trunk`: native VLAN differs between the two ends. (Seen transiently in this build, Section 7)
R1 subinterface down or wrong encapsulation	One VLAN cannot reach its gateway; trunks look fine; same visible symptom as the assigned fault	`show ip interface brief` on R1	Subinterface down, or `show running-config` shows the wrong `dot1Q` tag or mask. Distinguished from the assigned fault by the trunk check passing
Access port in the wrong VLAN	One PC takes an address from the wrong pool, or cannot reach its own department	`ipconfig` against the expected pool	`show vlan brief` or `show interfaces fa0/x switchport`
Wrong subnet mask on a host	Reaches some local addresses but not others; off-subnet traffic fails	`ipconfig`: compare mask with the pool	Correct the static setting or pool mask
Missing default route on R1	All internal traffic fine; only external traffic fails; `tracert` stops at R1	`show ip route`: no "Gateway of last resort"	Restore `ip route 0.0.0.0 0.0.0.0 10.23.1.2`
DHCP relay missing or scope exhausted	New or renewing PCs get no valid lease; hosts with static or existing addresses are fine	`ipconfig` shows no lease from the pool	R1 `show running-config` for `ip helper-address`; DHCP-SRV pool and lease count

6.1 Change Request 1 built and tested

ITEM	BEFORE CR1	AFTER CR1 (AS BUILT)	CHANGED?
Warehouse staff / PCs	12	20 (WH-PC-01…20)	Yes — endpoints only
SW-WH access ports in VLAN 30	Fa0/2–Fa0/13	Fa0/2–Fa0/21	Yes — port range only
VLAN 30 subnet	`10.23.0.128/26`	`10.23.0.128/26`	No
Gateway / R1 subinterface	`10.23.0.129, Gi0/0.30`	`10.23.0.129, Gi0/0.30`	No
Trunks, routing, NAT	—	—	No
DHCP WH-POOL	50 leases	50 leases (20 used)	No

"No redesign" is defined here as no change to any VLAN boundary, subnet mask, gateway, routing or trunk configuration. CR1 meets that: it is purely a cabling and switch-port assignment change, which is the point of provisioning uniform `/26`s and 24-port switches up front. Figure 4 shows the 20 Warehouse ports live in VLAN 30.

6.2 Branch office design proposal — not built, not simulated

The brief says only that a branch *may* open within 18 months, with no headcount or structure. So the branch is delivered as a design that proves the reservation works, not as a build. It is a proposal: the branch's real requirements would be confirmed with the client first, and none of it has been configured or tested in Packet Tracer. **The idea: the branch mirrors HQ's layout shifted by +128 in the third octet.**

PURPOSE	HQ (BUILT)	BRANCH (PROPOSED)
VLAN 10 / 20 / 30 / 40 blocks	`10.23.0.0 / .64 / .128 / .192 (/26)`	`10.23.128.0 / .64 / .128 / .192 (/26)`
VLAN 99 management / native	`10.23.1.16/28`	`10.23.129.16/28`
Site summary route	`10.23.0.0/23`	`10.23.128.0/23`
Inter-site link (/30)	`10.23.1.4/30 — R1 .5, branch router .6; inside HQ's existing /23 summary`	

1. **Link.** R1's third Gigabit port (Gi0/2) is unused and shut (Figure 5), so it can carry the inter-site link with no new HQ hardware. In production that link would be a site-to-site VPN over the ISP or a leased line, depending on what the client will pay for; in Packet Tracer a direct Ethernet link would stand in for it.
2. **Routing.** Two static routes are enough for two sites: at HQ, `ip route 10.23.128.0 255.255.254.0 10.23.1.6`; at the branch, a default route to `10.23.1.5`. A routing protocol is not justified at this scale.
3. **What changes at HQ:** a cable to Gi0/2, that port's configuration, the one static route, and **one line added to the NAT ACL** (`access-list 1 permit 10.23.128.0 0.0.1.255`) if the branch is to reach the internet through HQ. **What does not change:** every HQ VLAN, subnet, gateway, trunk, DHCP pool and switch. The NAT line is a real, small HQ change, and it is stated here rather than hidden.
4. **Capacity.** `10.23.128.0/17` holds 64 blocks of `/23`, each as large as HQ's whole footprint, so further sites need no HQ change either.

The full proposal, with a verification plan for when the branch is built, is in `docs/11-branch-office-design-proposal.md`.

⚠ 7 ISSUES ENCOUNTERED & LESSONS

The real problems hit during implementation, recorded as they happened rather than tidied away.

ISSUE	HOW IT WAS DETECTED	RESOLUTION	WHAT IT SHOWS
Core port plan used Gi0/2–Gi0/5 on SW-CORE	Checking the plan against the 2960-24TT's actual ports before building: it has only two Gigabit ports	Access uplinks moved to Fa0/1–Fa0/4; R1 kept Gi0/1	A port-only change; the VLAN and addressing design was untouched
Like-device links cabled straight-through	Reviewing the build against the rubric's "correct cabling" criterion	ISP–R1 and the four SW-CORE uplinks recabled as crossover; baseline resaved	Correct cabling should not depend on Auto-MDIX
Native-VLAN-mismatch and spanning-tree "inconsistent VLAN" messages	Console logs on SW-CORE while the switches were configured one at a time	None needed: each cleared once both ends of the trunk had native VLAN 99	A trunk's two ends must agree on the native VLAN
Empty NAT translation table	`show ip nat translations` run minutes after a successful ping	None needed: PAT entries time out when idle; the ping is the proof	Check translations immediately after generating traffic
Wrong config pasted onto the core switch during the fault demo	The Admin control test failed three times in a row (25%, then 100%, then 100% loss), which contradicted the prediction that only Warehouse would fail. The core switch's prompt read `SW-WH#`; `show cdp neighbors` listed 5 neighbours, so it was physically SW-CORE. The access-port block from `sw-wh.txt` had turned three of its trunks (to SW-SAL, SW-WH, SW-SRV) into VLAN 30 access ports	Re-applied `sw-core.txt` in full, confirmed all trunks with `show interfaces trunk`, saved, then re-injected only the one-line fault and re-ran every symptom test	Confirming a device's identity with CDP, and refusing to accept a symptom that did not fit the design

Why the last one matters. The first symptom test did not match the prediction: Admin failed too. Instead of recording it as the result, the mismatch was treated as a second, unknown problem and run down. All symptoms in Section 5 come from the clean rerun. The failed Admin pings from the misconfigured period remain visible higher up in the original capture's scrollback; they belong to that period, not to the assigned fault. An early explanation of those failures as spanning tree reconverging was wrong; the cause was the pasted configuration, and the repository records carry the corrected account.

Practices adopted from these incidents. Open each device's CLI fresh and read the prompt before pasting anything. Never paste a file whose header does not match the device. Verify identity with CDP when a symptom does not fit.

8 LIMITATIONS & FUTURE WORK

An honest account of what the delivered design does not do. None of these were required by the brief, but a production network would address them.

LIMITATION	CONSEQUENCE	WHAT WOULD BE ADDED
Single router and single core switch	R1 or SW-CORE failing takes the site down; one trunk carries all inter-VLAN traffic	A second router with HSRP/VRRP, a redundant core, or an EtherChannel to the router
No inter-VLAN access control	Any VLAN can reach any other, including Servers; the assigned fault shows how much depends on a few lines of configuration	ACLs on R1's subinterfaces, e.g. restricting access to the file server by department
No Layer 2 port security or DHCP snooping	A rogue device or DHCP server on an access port would not be stopped	Port security, DHCP snooping and dynamic ARP inspection on the access switches
Dependence on one DHCP server	If DHCP-SRV is down, new and renewing clients get no lease	A second DHCP server or a failover pair; shorter dependency chain via router-hosted pools as backup
Management over SVIs, no secure remote access	No SSH, no banner, no password policy was configured in the lab	SSH with local or AAA accounts, enable secret, console and VTY hardening, and a management ACL
Guest Wi-Fi reserved, not built	The `10.23.1.32/28` block exists on paper only	A guest VLAN, wireless controller or access point, and an ACL isolating it from internal VLANs
Branch office designed, not built	Section 6.2 is a proposal; no branch behaviour was simulated	Build and test it once the client confirms the branch's requirements
Simulation limits	Packet Tracer cannot show real performance, failover timing or hardware behaviour; results show correctness, not capacity	Validate on real equipment before deployment

9 REFLECTION

What worked

Designing the addressing before touching Packet Tracer paid off twice. Making every department a uniform `/26` meant Change Request 1 was a cabling exercise rather than a redesign, and reserving the top half of the `/16` meant the branch-office constraint cost nothing at HQ. Writing each device's configuration as a reviewable file first, applying one device at a time and checking with a `show` command before moving on, kept the build understandable: when something looked wrong, the list of things that could have changed was short.

What went wrong, and what it taught

Three mistakes were the most useful part of the project. First, the original core port plan assumed five Gigabit ports that a 2960-24TT does not have; the lesson was to check a plan against the real device before building, not after. Second, I first cabled the like-device links straight-through; Packet Tracer would have accepted them, but checking the work against the "correct cabling" criterion showed that "it works" and "it is right" are different standards. Third, and most important, during the fault demonstration I pasted the Warehouse switch's configuration onto the core switch. The symptom did not match my prediction, because the Admin control test failed too. The easy path was to record that as a result. Instead I treated the mismatch as a second, unknown problem, found the core switch's prompt reading `SW-WH#`, confirmed with CDP that it was physically the core switch, restored it, and re-ran the demonstration cleanly. The early failures were at first put down to spanning tree reconverging; that explanation was wrong, and it was corrected once the real cause was clear. That episode taught me more about troubleshooting than the planned fault did: a control test is only valuable if you are willing to believe it when it contradicts you.

What I learned about the assigned challenge

The VLAN-pruning fault is hard because nothing looks broken. The link is up, the switch ports are green, local traffic works, and the person reporting it will say "the Warehouse network is down". Scoping first (what works, what does not, and for whom) removed most of the network from suspicion before I ran a single diagnostic command, and one comparison, the allowed list on four ports side by side, found the cause. Writing the fault catalogue in Section 5.9 also showed me that several different faults produce nearly the same symptom, so the first check has to be the one that tells them apart, not the one that is easiest.

What I would do differently

1. Run `show ip nat translations` immediately after the NAT ping, and capture a standalone Warehouse inter-VLAN ping as its own test.
2. During the fault, also run `show ip interface brief` on R1 to show that R1's Warehouse subinterface is up and rule it out directly rather than by inference.
3. Add a hostname check, `show running-config | include hostname`, before pasting any configuration, so the wrong-device mistake is impossible rather than merely caught.
4. Add the security and redundancy features in Section 8, starting with inter-VLAN ACLs and a second DHCP path, since they would reduce the blast radius of exactly the kind of one-line mistake this project simulated.

10 GITHUB PORTFOLIO

Repository: **tshimo-agri-network** — https://github.com/MALULEKE-KS/tshimo-agri-network

```
tshimo-agri-network/
  README.md                          overview, milestone status, document index
  docs/                              living design and implementation record
    01–05                            Milestone 1 design (requirements → change requests)
    06                               control record: locked decisions, change log
    07, 08, 09                       physical build design, execution map, build walkthrough
    10-test-results.md               the 12-test log
    11-branch-office-design-proposal.md
    12-final-video-script-and-defence-prep.md
  configs/                           IOS command file per device + server setup
  packet-tracer/                     baseline (unconfigured) and milestone-2 (configured) .pkt
  screenshots/                       20 evidence captures, numbered by test
  troubleshooting/                   advanced-fault-isolation.md
  submissions/
    milestone-1/                     frozen M1 package
    milestone-2/                     frozen M2 package
    final/                           frozen Final package: this report + .pkt
```

10.1 Commit history

DATE	COMMIT
28 Aug 2026	Add Milestone 1 design package: requirements, topology, addressing plan
28 Aug 2026	Rename repository references to tshimo-agri-network, add eFundi submission text
28 Aug 2026	Restructure repo into docs/submissions layout, rebuild topology diagrams
28 Aug 2026	Rework design language, enlarge topology diagrams, tighten copy
28 Aug 2026	Ignore local session config file
28 Aug 2026	Add signing date to Milestone 1 declaration page
2 Oct 2026	Add Milestone 2 execution map and Packet Tracer build walkthrough
2 Oct 2026	Draft IOS configurations for switches, R1 and ISP, plus server setup
2 Oct 2026	Add baseline Packet Tracer topology and physical build evidence
2 Oct 2026	Milestone 2 implementation, testing and fault-isolation evidence

The Final-stage commits (this report, the Final package, the branch-office proposal, the video script and the README update) follow in the repository history on GitHub.

MILESTONE TRACKER & SIGN-OFF

PROJECT TIMELINE

Commencement	14 Aug 2026	**Complete**
Milestone 1 — Design Review	28 Aug 2026	**Design package complete**
Milestone 2 — Implementation Review	2 Oct 2026	**Built, tested, fault demonstrated**
Final Submission	16 Oct 2026	**Report, project file, portfolio and video**

DECLARATION

I declare that this report, the network it describes and its accompanying documentation are my own work, produced for CMPG 325 at North-West University, Mahikeng campus, in accordance with the NWU AI Policy. I built and configured the network in Cisco Packet Tracer, ran every test and captured every screenshot shown in this document, and I understand and can defend every design decision, configuration and troubleshooting step described here.

Signature — Kurhula Success Maluleke

Date

APPENDIX — DEVICE CONFIGURATIONS

As applied in Packet Tracer

The exact command files pasted into each device's CLI, reproduced from `configs/`. Lines starting with `!` are comments that IOS ignores. Server and PC settings are made in Packet Tracer's GUI and are given in Section 3.4 and in `configs/servers.md`.

SW-CORE — configs/sw-core.txt

```
! ============================================================
! SW-CORE — Tshimo Agri Supplies HQ — Layer 2 distribution switch
! Cisco 2960-24TT
! Role: trunks R1 (Gi0/1) down to the four access switches (Fa0/1-
4).
!       Holds the full VLAN database since every VLAN passes through
here.
! Paste this block into SW-CORE's CLI after Gate B (physical build)
passes.
! ============================================================

enable
configure terminal
hostname SW-CORE
no ip domain-lookup

! --- VLAN database ---
vlan 10
 name ADMIN
vlan 20
 name SALES
vlan 30
 name WAREHOUSE
vlan 40
 name SERVERS
vlan 99
 name MGMT-NATIVE
exit

! --- Uplink to R1 (802.1Q trunk, native VLAN 99) ---
interface gigabitEthernet0/1
 description Uplink to R1 Gi0/0
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 10,20,30,40,99
 no shutdown
exit

! --- Trunks to the four access switches ---
interface range fastEthernet0/1 - 4
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 10,20,30,40,99
 no shutdown
exit

interface fastEthernet0/1
 description Trunk to SW-ADM Fa0/1
exit
interface fastEthernet0/2
 description Trunk to SW-SAL Fa0/1
exit
interface fastEthernet0/3
 description Trunk to SW-WH Fa0/1
exit
interface fastEthernet0/4
 description Trunk to SW-SRV Fa0/1
exit

! --- Management SVI on the native VLAN ---
interface vlan 99
 ip address 10.23.1.18 255.255.255.240
 no shutdown
exit

ip default-gateway 10.23.1.17

end
write memory

! ------------------------------------------------------------
! If Packet Tracer's 2960 rejects "switchport mode trunk" with an
! encapsulation error, add this line before it on each trunk
interface:
!   switchport trunk encapsulation dot1q
! (Not expected on a real 2960-24TT — it's 802.1Q-only — but PT
model
! behaviour can vary; report the exact error if you see one.)
! ------------------------------------------------------------

! Verify with:
!   show vlan brief
!   show interfaces trunk
!   show ip interface brief
```

SW-ADM — configs/sw-adm.txt

```
! ============================================================
! SW-ADM — Tshimo Agri Supplies HQ — Admin access switch
! Cisco 2960-24TT
! Role: 8 Admin PCs on VLAN 10, trunked back to SW-CORE Fa0/1.
! Paste this block into SW-ADM's CLI after Gate B (physical build)
passes.
! ============================================================

enable
configure terminal
hostname SW-ADM
no ip domain-lookup

! --- VLANs used locally (full DB lives on SW-CORE) ---
vlan 10
 name ADMIN
vlan 99
 name MGMT-NATIVE
exit

! --- Uplink to SW-CORE (802.1Q trunk, native VLAN 99) ---
interface fastEthernet0/1
 description Uplink to SW-CORE Fa0/1
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 10,20,30,40,99
 no shutdown
exit

! --- Access ports: 8 Admin PCs ---
interface range fastEthernet0/2 - 9
 switchport mode access
 switchport access vlan 10
 spanning-tree portfast
 no shutdown
exit

! --- Management SVI on the native VLAN ---
interface vlan 99
 ip address 10.23.1.19 255.255.255.240
 no shutdown
exit

ip default-gateway 10.23.1.17

end
write memory

! Verify with:
!   show vlan brief
!   show interfaces trunk
!   show ip interface brief
```

SW-SAL — configs/sw-sal.txt

```
! ============================================================
! SW-SAL — Tshimo Agri Supplies HQ — Sales access switch
! Cisco 2960-24TT
! Role: 12 Sales PCs on VLAN 20, trunked back to SW-CORE Fa0/2.
! Paste this block into SW-SAL's CLI after Gate B (physical build)
passes.
! ============================================================

enable
configure terminal
hostname SW-SAL
no ip domain-lookup
```

```
! --- VLANs used locally (full DB lives on SW-CORE) ---
vlan 20
 name SALES
vlan 99
 name MGMT-NATIVE
exit

! --- Uplink to SW-CORE (802.1Q trunk, native VLAN 99) ---
interface fastEthernet0/1
 description Uplink to SW-CORE Fa0/2
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 10,20,30,40,99
 no shutdown
exit

! --- Access ports: 12 Sales PCs ---
interface range fastEthernet0/2 - 13
 switchport mode access
 switchport access vlan 20
 spanning-tree portfast
 no shutdown
exit

! --- Management SVI on the native VLAN ---
interface vlan 99
 ip address 10.23.1.20 255.255.255.240
 no shutdown
exit

ip default-gateway 10.23.1.17

end
write memory

! Verify with:
!   show vlan brief
!   show interfaces trunk
!   show ip interface brief
```

SW-WH — configs/sw-wh.txt

```
! ============================================================
! SW-WH — Tshimo Agri Supplies HQ — Warehouse access switch
! Cisco 2960-24TT
! Role: 20 Warehouse PCs (12 existing + 8 CR1) on VLAN 30,
!       trunked back to SW-CORE Fa0/3.
! Paste this block into SW-WH's CLI after Gate B (physical build)
passes.
!
! NOTE: SW-CORE's Fa0/3 is the trunk port used for the assigned
! troubleshooting fault (VLAN 30 removed from its allowed list).
! This file configures the healthy baseline only — the fault is
! injected later, on SW-CORE, per troubleshooting/advanced-fault-
isolation.md.
! ============================================================

enable
configure terminal
hostname SW-WH
no ip domain-lookup

! --- VLANs used locally (full DB lives on SW-CORE) ---
vlan 30
 name WAREHOUSE
vlan 99
 name MGMT-NATIVE
exit

! --- Uplink to SW-CORE (802.1Q trunk, native VLAN 99) ---
interface fastEthernet0/1
 description Uplink to SW-CORE Fa0/3
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 10,20,30,40,99
 no shutdown
exit

! --- Access ports: 20 Warehouse PCs (includes CR1 growth) ---
interface range fastEthernet0/2 - 21
 switchport mode access
 switchport access vlan 30
 spanning-tree portfast
 no shutdown
exit

! --- Management SVI on the native VLAN ---
interface vlan 99
 ip address 10.23.1.21 255.255.255.240
 no shutdown
exit

ip default-gateway 10.23.1.17

end
write memory

! Verify with:
!   show vlan brief
!   show interfaces trunk
!   show ip interface brief
```

SW-SRV — configs/sw-srv.txt

```
! ============================================================
! SW-SRV — Tshimo Agri Supplies HQ — Server access switch
! Cisco 2960-24TT
! Role: DHCP-SRV, DNS-SRV, FILE-SRV on VLAN 40, trunked back to SW-
CORE Fa0/4.
! Paste this block into SW-SRV's CLI after Gate B (physical build)
passes.
! ============================================================

enable
configure terminal
hostname SW-SRV
no ip domain-lookup

! --- VLANs used locally (full DB lives on SW-CORE) ---
vlan 40
 name SERVERS
vlan 99
 name MGMT-NATIVE
exit

! --- Uplink to SW-CORE (802.1Q trunk, native VLAN 99) ---
interface fastEthernet0/1
 description Uplink to SW-CORE Fa0/4
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 10,20,30,40,99
 no shutdown
exit

! --- Access ports: DHCP-SRV, DNS-SRV, FILE-SRV ---
interface range fastEthernet0/2 - 4
 switchport mode access
 switchport access vlan 40
 spanning-tree portfast
 no shutdown
exit

! --- Management SVI on the native VLAN ---
interface vlan 99
 ip address 10.23.1.22 255.255.255.240
 no shutdown
exit

ip default-gateway 10.23.1.17

end
write memory

! Verify with:
!   show vlan brief
!   show interfaces trunk
!   show ip interface brief
```

R1 — configs/r1.txt

```
! ============================================================
! R1 — Tshimo Agri Supplies HQ — edge router
! Cisco 2911
! Role: router-on-a-stick inter-VLAN routing on Gi0/0 (trunk to SW-
CORE Gi0/1),
!       DHCP relay to DHCP-SRV, WAN to ISP on Gi0/1, NAT/PAT
overload.
! Paste AFTER the five switch configs are in.
! ============================================================

enable
configure terminal
hostname R1
no ip domain-lookup

! --- Physical trunk interface: no IP, just bring it up ---
interface gigabitEthernet0/0
```

```
 description Trunk to SW-CORE Gi0/1
 no ip address
 no shutdown
exit

! --- One subinterface per VLAN = that VLAN's default gateway ---
interface gigabitEthernet0/0.10
 description VLAN 10 ADMIN gateway
 encapsulation dot1Q 10
 ip address 10.23.0.1 255.255.255.192
 ip helper-address 10.23.0.194
 ip nat inside
exit

interface gigabitEthernet0/0.20
 description VLAN 20 SALES gateway
 encapsulation dot1Q 20
 ip address 10.23.0.65 255.255.255.192
 ip helper-address 10.23.0.194
 ip nat inside
exit

interface gigabitEthernet0/0.30
 description VLAN 30 WAREHOUSE gateway
 encapsulation dot1Q 30
 ip address 10.23.0.129 255.255.255.192
 ip helper-address 10.23.0.194
 ip nat inside
exit

interface gigabitEthernet0/0.40
 description VLAN 40 SERVERS gateway
 encapsulation dot1Q 40
 ip address 10.23.0.193 255.255.255.192
 ip nat inside
exit

! Native VLAN 99: untagged frames on the trunk belong here
interface gigabitEthernet0/0.99
 description VLAN 99 MGMT-NATIVE gateway
 encapsulation dot1Q 99 native
 ip address 10.23.1.17 255.255.255.240
exit

! --- WAN to ISP ---
interface gigabitEthernet0/1
 description WAN to ISP
 ip address 10.23.1.1 255.255.255.252
 ip nat outside
 no shutdown
exit

! --- Default route out to the ISP ---
ip route 0.0.0.0 0.0.0.0 10.23.1.2

! --- NAT/PAT: translate the business/server range (10.23.0.0/24)
only.
!     WAN and management (10.23.1.0/24) are deliberately excluded. -
--
access-list 1 permit 10.23.0.0 0.0.0.255
ip nat inside source list 1 interface gigabitEthernet0/1 overload

end
write memory

! Verify with:
!   show ip interface brief        (all subinterfaces + Gi0/1 up/up)
!   show ip route                  (C routes for each VLAN, S*
default to 10.23.1.2)
!   show ip nat translations       (after a PC pings 203.0.113.1)
```

ISP — configs/isp.txt

```
! ============================================================
! ISP — simulated upstream provider
! Cisco 2911
! Role: far end of the R1 WAN /30, plus a loopback acting as an
!       "internet" test target for NAT verification.
!
! ISP port facing R1 Gi0/1 is Gi0/0. Confirmed 2 Oct 2026: this file
was
! applied unchanged, R1 Gi0/1 came up/up, and a PC reached
203.0.113.1
! through NAT — impossible unless the link lands on the addressed
Gi0/0.
! ============================================================

enable
configure terminal
hostname ISP
no ip domain-lookup

interface gigabitEthernet0/0
 description WAN to Tshimo R1 Gi0/1
 ip address 10.23.1.2 255.255.255.252
 no shutdown
exit

! Simulated internet host
interface loopback0
 description Internet test target
 ip address 203.0.113.1 255.255.255.255
exit

end
write memory

! No route back to 10.23.0.0/24 is needed: R1 NATs every inside
! host to 10.23.1.1, which the ISP already reaches directly.
```